CSES Sliding Window Problems — Problem Reference

All tasks: 1 s, 512 MiB. Windows are processed from left to right.

1076 — Sliding Window Median

For every length-k window of $x_1, \ldots, x_n$, find its median. If k is even, use the smaller of the two middle elements. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ medians

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

1077 — Sliding Window Cost

For every length-k window, find the minimum total cost to make all elements equal, where changing u to v costs $|u-v|$. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ minimum costs

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3219 — Sliding Window Mex

For every length-k window, find its mex: the smallest nonnegative integer not present in the window. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ mex values

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $0 \le x_i \le 10^9$.

3220 — Sliding Window Sum

The array is generated online. For every length-k window compute its sum, then xor all window sums together. **In:** `n k`; `x a b c`

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.

Out: xor of all window sums

Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$. Use 64-bit sums.

3221 — Sliding Window Minimum

The array is generated online. For every length-k window compute its minimum, then xor all window minimums together. **In:** `n k`; `x a b c`

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.

Out: xor of all window minimums

Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$.

3222 — Sliding Window Distinct Values

For every length-k window, count how many distinct values it contains. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ distinct-value counts

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

3223 — Sliding Window Inversions

For every length-k window, count inversions: pairs of positions $p < q$ inside the window with $x_p > x_q$. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ inversion counts

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3224 — Sliding Window Mode

For every length-k window, output its most frequent value. If several values have maximum frequency, choose the smallest. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ modes

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$.

3227 — Sliding Window Advertisement

A fence has n unit-width boards of heights x_i. For every k consecutive boards, find the largest axis-aligned rectangle that fits under those bars. **In:** `n k`; $x_1 \ \ldots \ x_n$

Out: $n-k+1$ maximum rectangle areas

Constraints: $1 \le k \le n \le 2 \cdot 10^5$, $1 \le x_i \le 10^9$. Use 64-bit.

3405 — Sliding Window Or

The array is generated online. For every length-k window compute the bitwise OR of its elements, then xor all window ORs together. **In:** `n k`; `x a b c`

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.

Out: xor of all window ORs

Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$.

3426 — Sliding Window Xor

The array is generated online. For every length-k window compute the bitwise XOR of its elements, then xor all window XORs together. **In:** `n k`; `x a b c`

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \le i \le n$.

Out: xor of all window XORs

Constraints: $1 \le k \le n \le 10^7$, $0 \le x, a, b \le 10^9$, $1 \le c \le 10^9$.